



COLLEGE
OF
MEDICINE

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Diagnosis of skin disorders
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Objectives

- How to reach the diagnosis of skin diseases.
- Highlight on types of skin lesions.
- Defining the description of skin lesions, their configuration and arrangements.
- What are the tools needed to examine the skin.
- Knowledge about investigations in dermatology.
- What are patch and prick tests.

Diagnosis of skin disorders based on:

- History
- Examination
- Investigations

Examination:

- The lightening must be uniform and bright, daylight is best.
- Good exposure needed.
- Make up must be removed.
- Negative signs are important as positive signs, like in photodermatosis.
- Examination for diagnosis is based on the **distribution** of lesions and on their **morphology** and **configuration**.

Distribution:

- *Site of the lesions.
- *See if the lesions localized, generalized, dermatomal, universal. *uni or bilateral*
- *Symmetrically distributed or asymmetrical.
bilateral Symmetry implies a systemic origin, whereas unilaterality or asymmetry implies an external cause.
- *Examine mouth, hair and nails. *part of skin*

Morphology

Terminology of skin lesions:

- Primary lesions *↪ if lesion remain unchanged (constant)*
- Secondary lesions *↪ if it evolve from the primary lesions*

Primary lesions:

Erythema is redness caused by vascular dilatation. *blanch under pressure*



Macule is a small flat area of altered color or texture less than 0.5 cm.



Patch is like macule greater than 0.5 cm



Papule is a small ^{soft tissue} solid elevation of skin, less than 0.5 cm in diameter.



Similar in size to papule
→ soft tissue

Papilloma is a nipple-like projection from the skin.

Nodule is a solid mass in the skin, usually greater than 0.5 cm in diameter, in both width and depth

↳
but less than 2cm



Plaque is an elevated area of skin greater than 2 cm in diameter but without substantial depth.



Tumor is an enlargement of the tissues by normal or pathological material or cells that form a mass, usually more than 2 cm in diameter. ^{and deeply} Because the word ‘tumor’ can scare patients, tumors may courteously be called ‘large nodules’, especially if they are not malignant.



Vesicle is a circumscribed elevation of skin, less than 0.5 cm in diameter, and containing fluid.



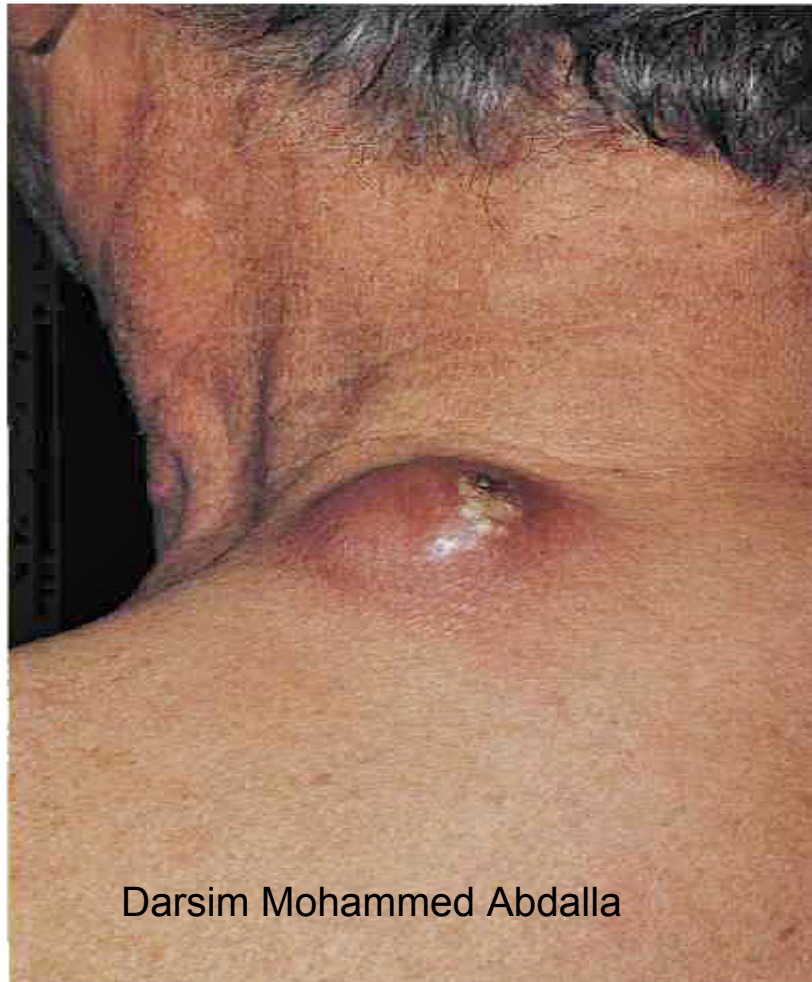
Bulla is a circumscribed elevation of skin over 0.5 cm in diameter and containing fluid.



Pustule is a visible accumulation of pus in the skin less than 0.5 cm.



Abscess is a localized collection of pus in a cavity, more than 0.5 cm in diameter.



Wheal is an elevated white compressible evanescent area produced by dermal edema. It is often surrounded by a red axon-mediated flare. Although usually less than 2 cm in diameter, some wheals are huge.

Angioedema is a diffuse swelling caused by edema extending to the subcutaneous tissue..

↳ not red, no itching, usually around eye, mouth, nose

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Petechiae are pinhead-sized macules of blood in the skin.

The term **purpura** describes a larger macule or papule of blood in the skin. Such blood-filled lesions do not blanch if a glass lens is pushed against them (diascopy).



Ecchymosis is a larger extravasation of blood into the skin.

Hematoma is a swelling from gross bleeding.

muscle bleeding



Telangiectasia is the visible dilatation of small cutaneous blood vessels. (capillaries)



Poikiloderma is a combination of atrophy, reticulate hyperpigmentation and telangiectasia.



Secondary lesions:

These evolve from primary lesions.

^{كبريتات}
Scale is a flake arising from the
horny layer._{root of epidermis}



^{قشعرير}
Crust may look like a scale, but is
composed of dried blood or tissue
fluid.

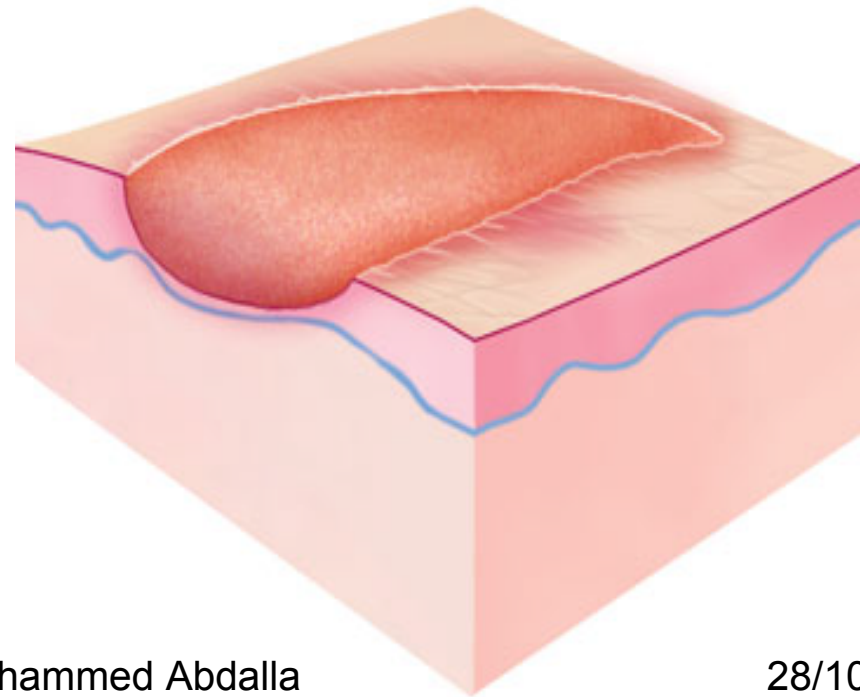


Keratosis is a horn-like thickening of the stratum corneum.

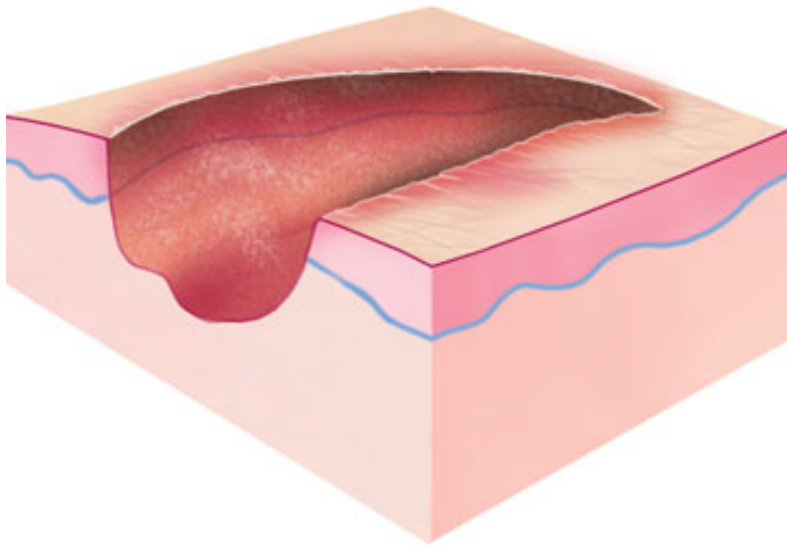


Ablasion is similar to erosion

Erosion is an area of skin denuded by a complete or partial loss of only the epidermis. Erosions heal without scarring.



Ulcer is an area of skin from which the whole of the epidermis and at least the upper part of the dermis has been lost. Ulcers may extend into subcutaneous fat, and heal with scarring. *even if not involve hypodermis*



by severe scratching
An **excoriation** is an ulcer or erosion produced by scratching.

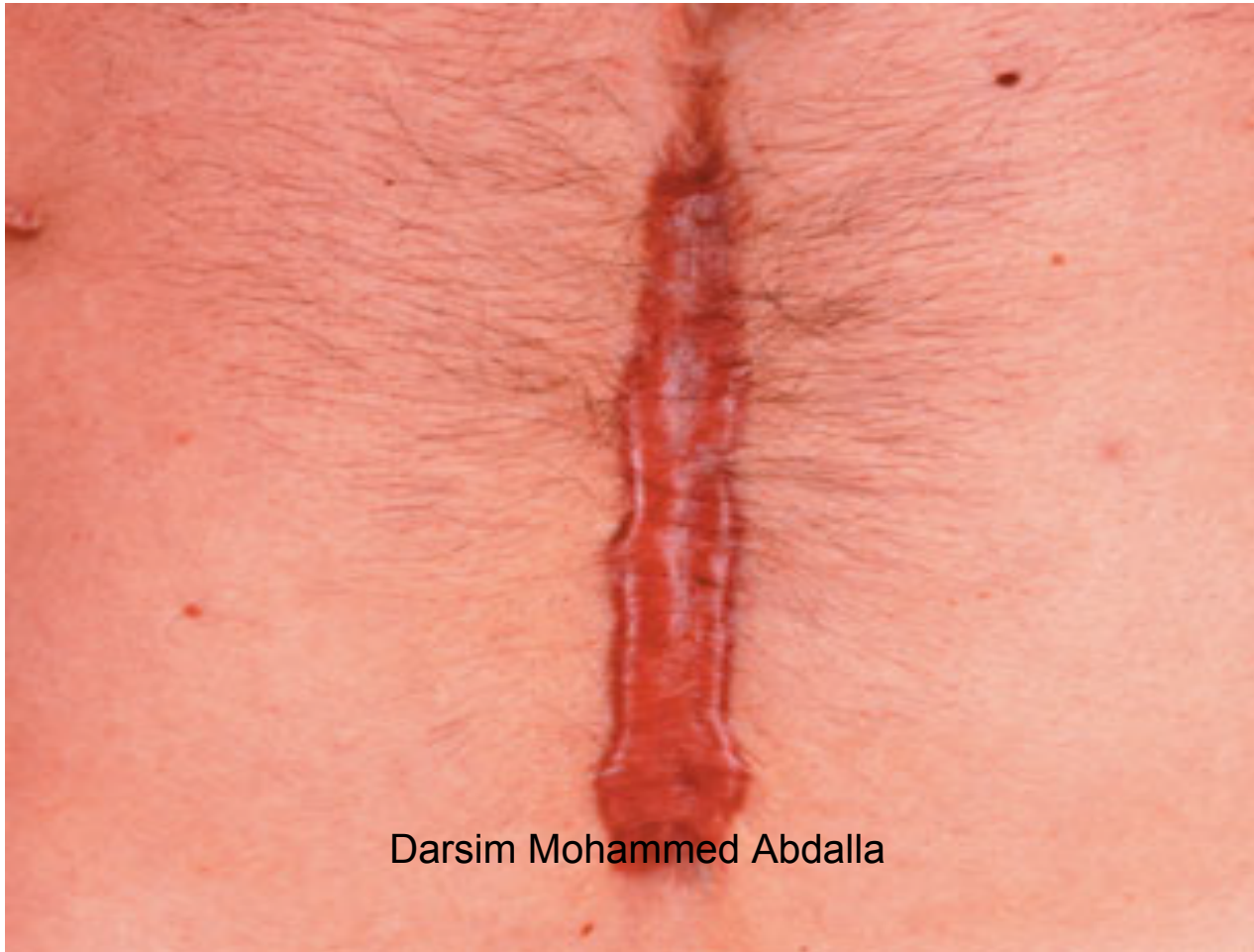
Sinus is a cavity or channel that permits the escape of pus or fluid.



Fissure is a slit in the skin.



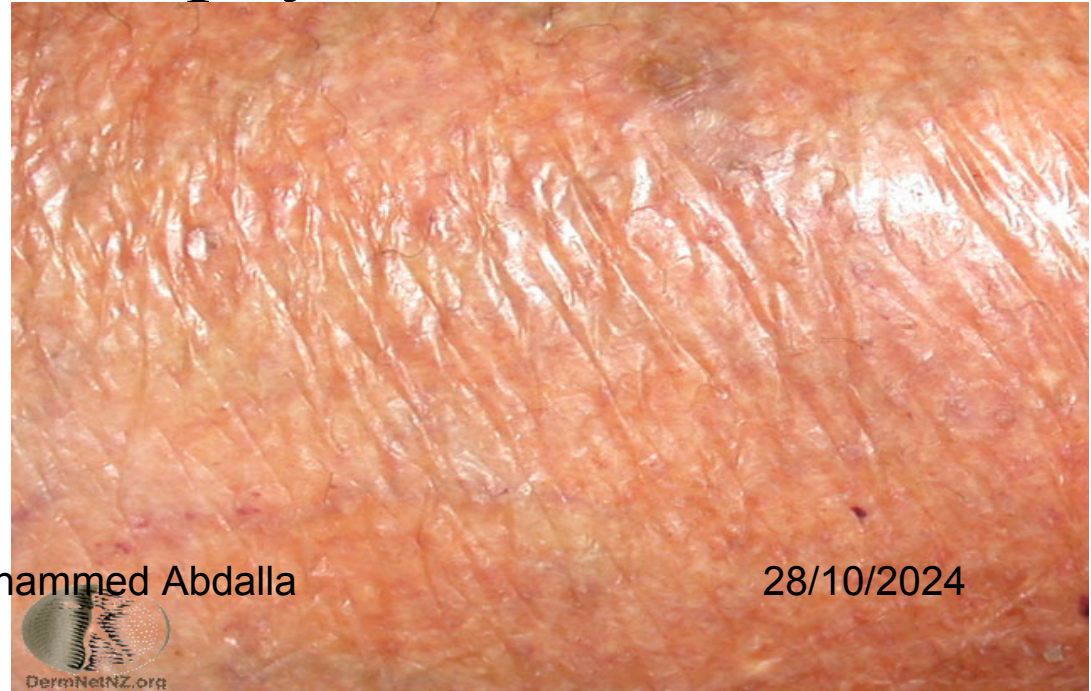
Scar is a result of healing, where normal structures are ^{wound} permanently replaced by fibrous tissue.



Atrophy is a thinning of skin caused by diminution of the epidermis, dermis or subcutaneous fat. When the epidermis is atrophic it may crinkle like cigarette paper, appear thin and translucent, and lose normal surface markings. Blood vessels may be easy to see in both epidermal and dermal atrophy

can be caused by steroid

Not necessarily happening together



excessive scratch
Lichenification is an area of thickened skin with increased markings.



Stria (stretch mark) is a streak-like linear atrophic pink, purple or white lesion of the skin caused by changes in the connective tissue.



Pigmentation either more or less than surrounding skin, can develop after lesions heal.

Description of lesions:

- Color: eg. Salmon-pink, lilac, violet.
- Sharpness of edge: e.g.: well defined, ill defined.
- Surface contour: dome-shaped, umblicated, spire like.
- Geometric shape: nummular, oval, irregular.
- Texture: rough, silky, smooth, hard.
- Temperature: hot, warm.

Special adjectives which warrant definition:

Nummular: round or coin shape. *→ lesion in edge and center*

Annular: ring-like. *→ edge is lesion, but center is clear*

Circinate: circular.

Arcuate: curved. *→ half circle*

Discoid: disc-like.

Gyrate: wave-like.

Retiform and reticulate: net-like.

Targetoid: target-like or bull's eye.

Polycyclic: formed from coalescing circles, or incomplete rings.

Configuration:

After describing primary and secondary lesions, look for arrangement and configuration that can be, for e.g. discrete, confluent, grouped, linear, annular, nummular, serpiginous, arcuate, segmental and dermatomal.

Investigations:

1. Special tools and techniques: *if you click here*

***Magnifying lens:**

Subtle changes in the skin become more apparent when enlarges.



***Wood's light:**

Emitting long wavelength ultraviolet radiation.

-Fluorescence is seen in some fungal infections, erythrasma and pseudomonas infection.



- Some subtle disorders of pigmentation can be seen more clearly under wood's light, e.g. the pale patches of tuberous sclerosis, low-grade vitiligo and pityriasis versicolor, and the darker café au lait patches of tuberous sclerosis.
- The urine of porphyria cutanea tarda fluoresces coral pink.

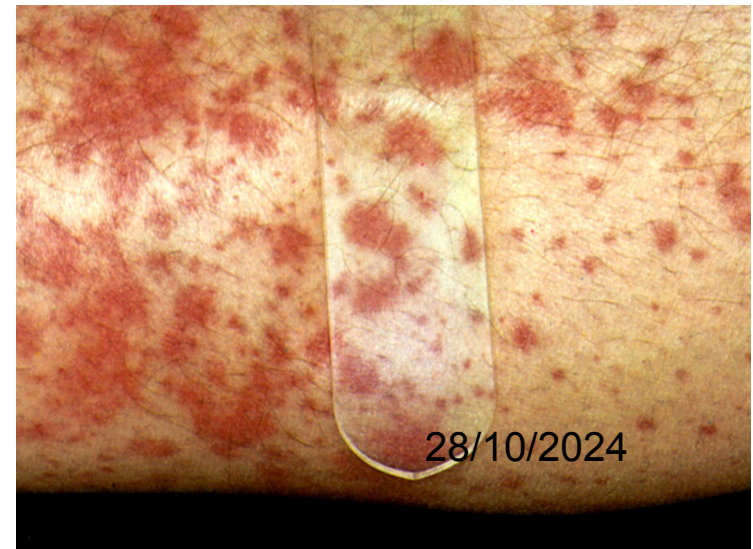


***Diascopy:**

A glass slide is pressed on vascular lesions to blanch them and verify that their redness is caused by vasodilatation and to unmask their underlying color.

-Confirm the presence of extravasated blood in the dermis (i.e. petechia and purpura, the appearance of which do not change on pressure)

-Granulomatous condition like sarcoid reveal a color of apple jelly.



*Dermatoscopy (dermoscopy):

Non-invasive technique, used as aid in the diagnosis of pigmented lesions (benign and malignant); visualization of scabies mites in their burrows; and in deciding if alopecia is scarring or not



*Photography:

Helps to record the baseline appearance of a lesion or rash, so that changes can be assessed objectively at

2. Side-room and office tests:

***Potassium hydroxide preparations for fungal infections:**

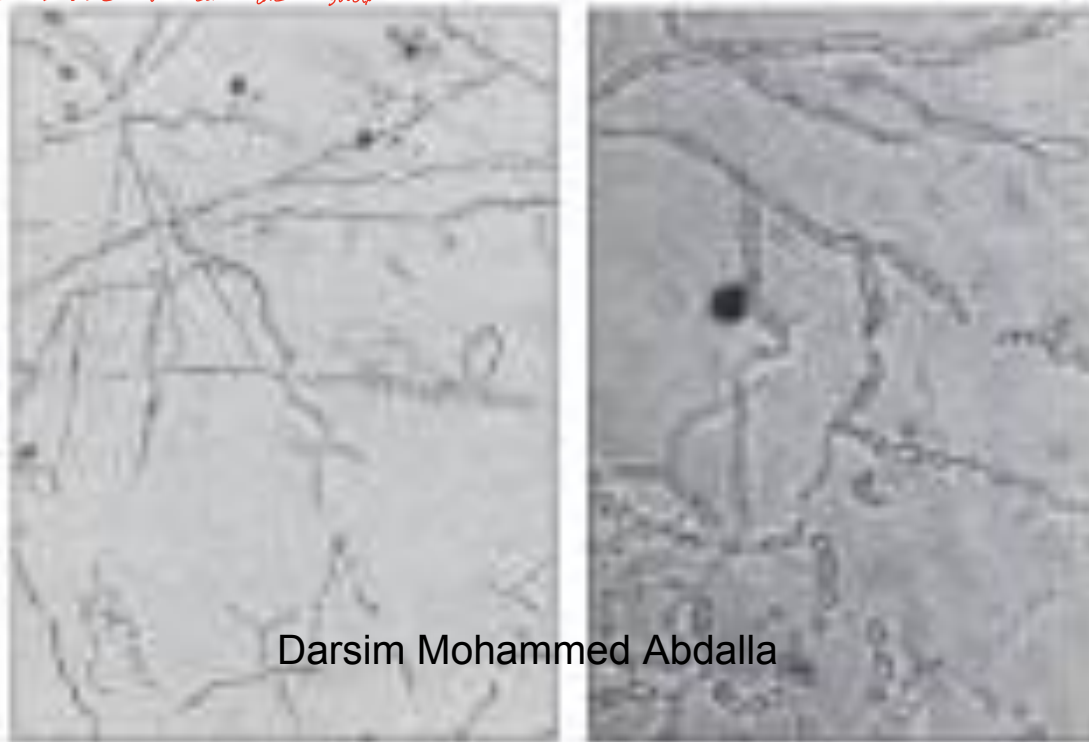
If fungal infection is suspected, scales or plucked hairs are dissolved in an aqueous solution of 20% KOH containing 40% dimethyl sulfoxide (DMSO).

The scale from the edge of a scaling lesion is vigorously scraped on to a glass slide with scalpel blade or the edge of a second glass slide.

Other samples can include nail clippings, the roofs of blisters, hair pluckings and contents of pustules when a candida infection is suspected.

A drop or two of the KOH solution is run under the coverslip. After 5-10 minutes the mount is examined under microscope, fungal and candidal hyphae can be readily detected.

No heat is required if DMSO is included in the KOH solution.



***Bacterial swab:**

If a wound or rash is smelly, weeping, crusting or has a yellow (*Staphylococcus aureus*) or green (*Pseudomonas*) discoloration, it may be worth while taking swab for bacterial culture and sensitivity.



Caused by *Sarcoptes scabiei* (Parasite)

*Detection of scabies mite:

Burrows in an itchy patient are diagnostic of scabies.

The burrow should be examined under a magnifying glass or dermatoscope, the aracus is seen as a tiny black or gray dot at the most recent, least scaly end.



It can be removed by a sterile needle and placed on a slide and seen under microscope.

Alternatively if mites can not be seen, possible burrows can be vigorously scraped with a scalpel blade, moistened with liquid paraffin and the scrapings transferred to a slide.

→ used between fingers, forearm, around wrist



***Cytology (Tzank smear):**

Cytology can aid the diagnosis of viral infections such as herpes simplex and zoster, and of bullous diseases such as pemphigus.

A blister roof is removed and the cells from the base of the blister are scrapped with a blade.

These cells are smeared on to a microscope slide, air-dried and fixed with methanol.

They are then stained with Gimsa, toluidine blue or Wright's stain.

Acantholytic cells are seen in pemphigus and multinucleate giant cells are diagnostic for herpes infections.

*Patch tests:

- Used for detecting the allergens responsible for allergic contact dermatitis.
 - The test detects type ~~VI~~^{IV} delayed hypersensitivity reactions. *→ for allergic → for delayed type IV*
 - It involves applying a chemical to the skin and then watching for dermatitis to develop 48-96 hours later.
- Either suspected individual antigens, or a battery of antigens that are common culprits, can be tested.

The test materials are applied to the back under aluminum discs or patches.

The patches are left in place for 48 hours, and then after careful marking, are removed.

The sites are inspected 10 minutes later, again 2 days later, and sometimes even later if doubtful reactions require further assessment.



The readings are scored according to the reaction seen:

NT not tested

- No reaction

± Doubtful reaction (minimal erythema)

+ Weak positive reaction (erythema and may be papules).

++ Strong reaction (palpable erythema and/or vesicles).

+++ Extreme reaction (intense palpable erythema, coalescing vesicles and/or bullae).



*Prick testing:

- Is much less helpful in dermatology.
- It detects immediate (type I hypersensitivity like urticaria, and patients should not have taken systemic antihistamines for at least 48 hours before the test.
- Commercially prepared diluted antigens and a control ^{normal saline} are placed as single drops on marked areas of the forearm.

- The skin is gently pricked through the drops using separate sterile fine needles. The prick should not cause bleeding. The drops are then removed with a tissue wipe.
- After 10 minutes the sites are inspected and the diameter of any wheal measured and recorded.

A result is considered positive if the test antigen causes a wheal of 4 mm or greater and the control elicits a negative reaction.

*Skin biopsy:

A piece of tissue is removed surgically for histological examination and, sometimes, for other tests (e.g. culture for organisms).

Skin biopsy may be incisional, when just part of a lesion is removed, or excisional, when the whole lesion is cut out which is preferable for most small lesions (up to 0.5 cm in diameter) but incisional biopsy is chosen for larger lesions and should include a piece of the surrounding normal tissue. *mark it two (this normal, this diseased)*

-Local anaesthetic;

Lidocaine 1-2% is used. Sometimes, epinephrine 1:200000 is added, this causes vasoconstriction and reduced clearance of the local anaesthetic and prolongation of the local anaesthetic effect.

Plain lidocaine ^{with epinephrine} should be used on the fingers, toes and the penis as the prolonged vasoconstriction produced by epinephrine can be dangerous here. ^{leading to gangrene}

Epinephrine is also best avoided in diabetics with small vessel disease, in those with history of heart disease (including dysrhythmias).

In patients taking non-selective α -blockers and tricyclic antidepressants (because of potential interaction) and in uncontrolled hyperthyroidism, it is wise to avoid local anaesthesia during early pregnancy.

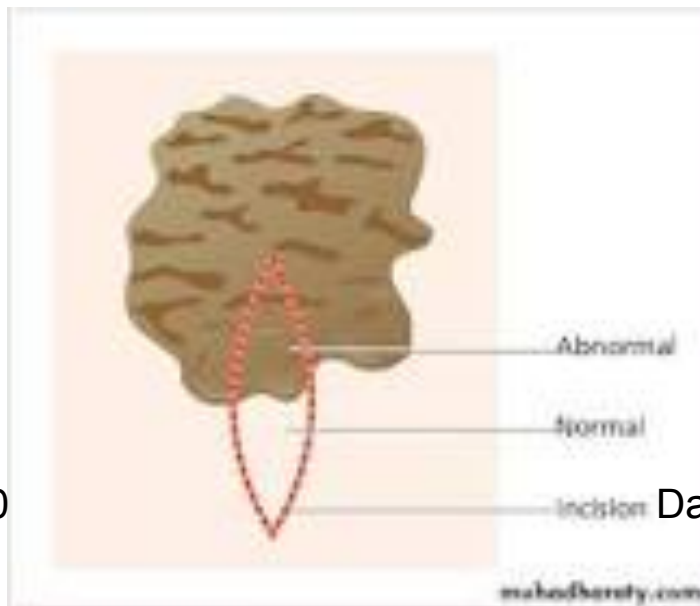
↳ Contraindication

If the local anaesthetic is injected in to the subcutaneous fat, it will be relatively pain-free, will produce a diffuse swelling of the skin and will take several minutes to induce anaesthesia. Intradermal^(in dermis) injections are painful and produce a discrete wheal associated with rapid anaesthesia.

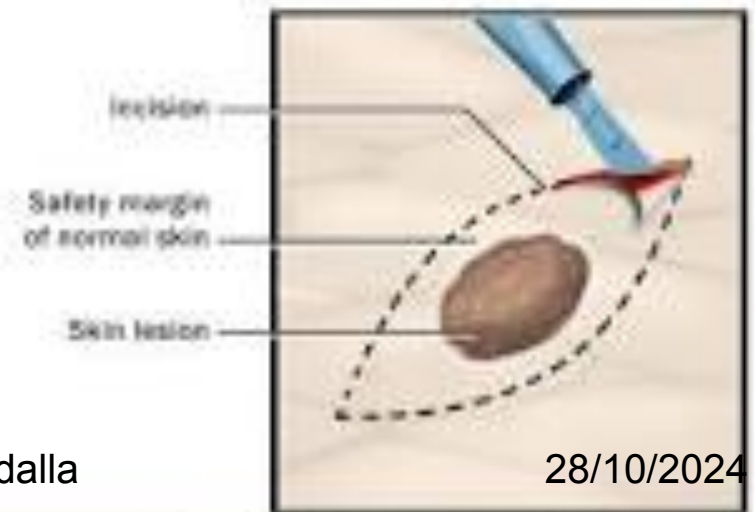
-Scalpel biopsy:

This provides more tissue than a punch biopsy, for obtaining specimens with both normal and abnormal skin for comparison and for removing small lesion in toto (excision biopsy).

An elliptical piece of skin is excised, the specimen should include the subcutaneous fat.



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- The wound is then sutured; non-absorbable 3/0 sutures are used for biopsies on the legs and back, 5/0 for the face and 4/0 for elsewhere.
- Stitches are usually removed from the face in 4-7 days, from the anterior trunk and arms in 7-10 days and from the back and legs in 10-14 days.

-Punch biopsy: used for incisional or small excisional

The skin is sampled with a small (3-4mm diameter) tissue punch.

A cylinder of skin is incised with the punch by rotating it back and forth. Skin is lifted up carefully with a needle or forceps and the base is cut off at the level of subcutaneous fat, the defect is sutured.



***Laboratory tests:**

Include various assess of blood, serum and urine, bacterial, fungal and viral culture from skin and other specimens, immunofluorescent and immunohistologic examinations, radiography and ultrasonograhly.

